

Midterm Exam

Version A

Friday, October 9, 2026 · in class

Name: _____ Student ID: _____

40 points | Suggested working time 40 minutes

Use a basic calculator. Show your working. Allow about 7, 10, 7, 10 and 3 minutes for Sections 1–5, plus 3 minutes to read and check. Sections 1 and 5 are marked on clear reasoning, not on matching one “right” answer. The data are synthetic teaching data from a fictional seaside café.

Section	What it covers	Points	Score
1	Choose the inputs	8	
2	Read the LINEST output	12	
3	Errors and outliers	8	
4	Did the price change sales?	9	
5	Summary for the owner	3	
Total		40	

SECTION 1 · Choose the inputs

8 points · about 7 minutes

A seaside café wants to predict **tomorrow's gelato sales** (cups) **before it opens**. Table 1 lists information it could use. Across 168 days of data, **temperature and sunshine are strongly related to each other ($r = 0.78$)**. Their correlations with gelato sales are 0.58 (temperature) and 0.49 (sunshine).

Table 1. Candidate information.

#	Information
1	Tomorrow's temperature forecast (°C)
2	Tomorrow's sunshine forecast (hours)
3	Whether tomorrow is a weekend (yes or no)
4	Whether a beach event is on the calendar tomorrow (yes or no)
5	Tomorrow's planned gelato price (€)
6	Tomorrow's iced coffee sales (cups), counted at closing time

1a Choose **three** inputs for the model. For each, say whether it should push gelato sales **up or down**, and give a one-line reason. 3 pts

Input (name or number)	Up or down?	Reason (one line)

1b One candidate in Table 1 is **unsuitable** for this forecast. Which one, and what is the problem? 2 pts

1c Should temperature **and** sunshine both go into the same model? Explain, and say which one you would keep. 2 pts

1d How would you check the model's prediction errors on days that were **not** used to fit it? 1 pt

SECTION 2 · Read the LINEST output

12 points · about 10 minutes

A model of daily gelato sales (cups) was fitted to 168 days with $\text{=LINEST}(y, X, \text{TRUE}, \text{TRUE})$. The three X columns were entered in this order: X_1 = temperature (°C), X_2 = sunshine (hours), X_3 = beach event (1 = event, 0 = no event). Use Table 2 to read Table 3.

Table 2. How LINEST lays out its results.

Col A (X_3)	Col B (X_2)	Col C (X_1)	Col D (Intercept)	Statistic description
Slope m_3	Slope m_2	Slope m_1	Intercept (b)	Coefficients for: $y = m_1X_1 + m_2X_2 + m_3X_3 + b$
SE – m_3	SE – m_2	SE – m_1	SE – Intercept	Standard error for each coefficient
R^2	SE – Y	Blank	Blank	Quality of the model fit
F-statistic	df	Blank	Blank	Analysis of variance metrics
SS – Reg	SS – Resid	Blank	Blank	Sum of squares metrics

Table 3. LINEST output for the model (shaded cells are blank).

Col A	Col B	Col C	Col D
27.09	1.99	4.31	72.87
5.18	1.37	1.01	16.36
0.43	27.22		
41.82	164		
92995.94	121553.47		

2a Write the fitted equation, using the numbers in Table 3 and the variable names.

2 pts

2b Each statement below misreads the output. Rewrite each one correctly, keeping the units and conditions.

3 pts

(i)	"The intercept means the café sells 72.87 cups on an average day."
(ii)	"Beach events cause the café to sell 27.09 more cups."
(iii)	" $R^2 = 0.43$ means the model's predictions are correct 43% of the time."

2c A coefficient is clearly different from zero if it is **more than 2 standard errors** away from zero. (a) Which one predictor is **not**? Show the comparison. (b) Use Section 1 to suggest why. (c) What would you do with this predictor?

3 pts

2d Tomorrow is forecast to be 26 °C with 8 hours of sunshine and a beach event. (a) Predict gelato sales. (b) Use SE – Y to give a rough range: prediction $\pm 2 \times$ SE – Y. (c) Actual sales turn out to be **150 cups**. Is that surprising? Explain.

4 pts

SECTION 3 · Errors and outliers

8 points · about 7 minutes

3a · Errors. Sales and errors are in cups. These are constructed exercise numbers, not workbook output. **Residual = actual – predicted.** **MAE = (sum of the absolute residuals) ÷ n.**

Day	Actual	Predicted	Residual
Mon	180	190	
Sat		230	+15
Wed	160		–5

3a-i Fill the blanks in the table.

1 pt

3a-ii Calculate the MAE of these predictions.

1 pt

3a-iii Model Z has residuals of –6, –6, –6 on the same three days. Which model has the lower MAE? What do the identical signs tell you about Model Z?

2 pts

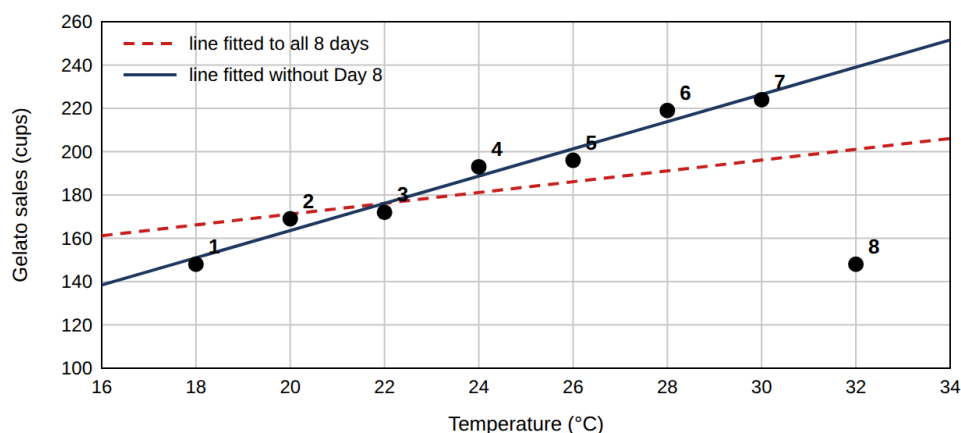
3b · Outliers. Table 4 and Figure 1 show gelato sales on 8 days (constructed example). Table 5 gives the straight-line results with and without Day 8.

Day	1	2	3	4	5	6	7	8
Temperature (°C)	18	20	22	24	26	28	30	32
Sales (cups)	148	169	172	193	196	219	224	148

Table 5. Results for the straight line (sales against temperature).

	All 8 days	Without Day 8
Slope (cups per °C)	2.49	6.29
R ²	0.17	0.97
SE – Y (cups)	28.77	5.18

Figure 1. The 8 days, with the line fitted to all of them (dashed) and the line fitted without Day 8 (solid).



3b-i Which day does not follow the pattern of the others?

1 pt

3b-ii Compare the results with and without that day. What impact does one day have on the slope and on R²?

2 pts

3b-iii Before deleting it, what would you do?

1 pt

SECTION 4 · Did the price change sales?

9 points · about 10 minutes

On 30 June the café raised the gelato price from **€4.00** to **€4.50**. Table 6 shows gelato sales (cups) on 5 weekdays **before** (Group 1) and 5 weekdays **after** (Group 2). Both samples are weekdays with similar weather (about 24 °C).

Formulas. $s = \sqrt{[\sum (x - \bar{x})^2 \div (n - 1)]}$ · $s_p = \sqrt{[(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] \div (n_1 + n_2 - 2)}$ · $t = (\bar{x}_1 - \bar{x}_2) \div [s_p \times \sqrt{(1/n_1 + 1/n_2)}]$ · $df = n_1 + n_2 - 2$

For $df = 8$ the two-tailed 5% critical value is 2.31. A negative t only means that Group 2 has the larger mean.

Table 6. Worksheet (fill in the blank cells).

Day	Before (x_1)	$x_1 - \bar{x}_1$	$(x_1 - \bar{x}_1)^2$	After (x_2)	$x_2 - \bar{x}_2$	$(x_2 - \bar{x}_2)^2$
1	196			184		
2	204			192		
3	188			176		
4	212			196		
5	200			182		
Sum						
Mean $\bar{x}$						
s						

4a Complete Table 6: the two means, the two sums of squared deviations, and s for each group (2 decimals). **3 pts**

4b Calculate s_p and t (2 decimals). **2 pts**

Sheets gives **=T.TEST(B2:B6, C2:C6, 2, 2) = 0.031** (two-tailed, equal variances).

4c (i) State the null hypothesis in words. (ii) Is the difference statistically significant at 5%? Explain using the p -value and your t . (iii) In plain words, what does $p = 0.031$ mean? **3 pts**

4d Name one other change, besides the price, that could explain the drop in sales. **1 pt**

SECTION 5 · Summary for the owner

3 points · about 3 minutes

5 Write 3–4 sentences to the café owner, who has no statistics background. Include: (i) which inputs you recommend for forecasting gelato sales and the main finding from the LINEST output; (ii) what the price test showed; (iii) one caution. **3 pts**

— End of exam —